DAY 7 OF 7

# Caching, charts & final review

Faster APIs with Redis caching, summary charts for admin and user, and a project ready to submit

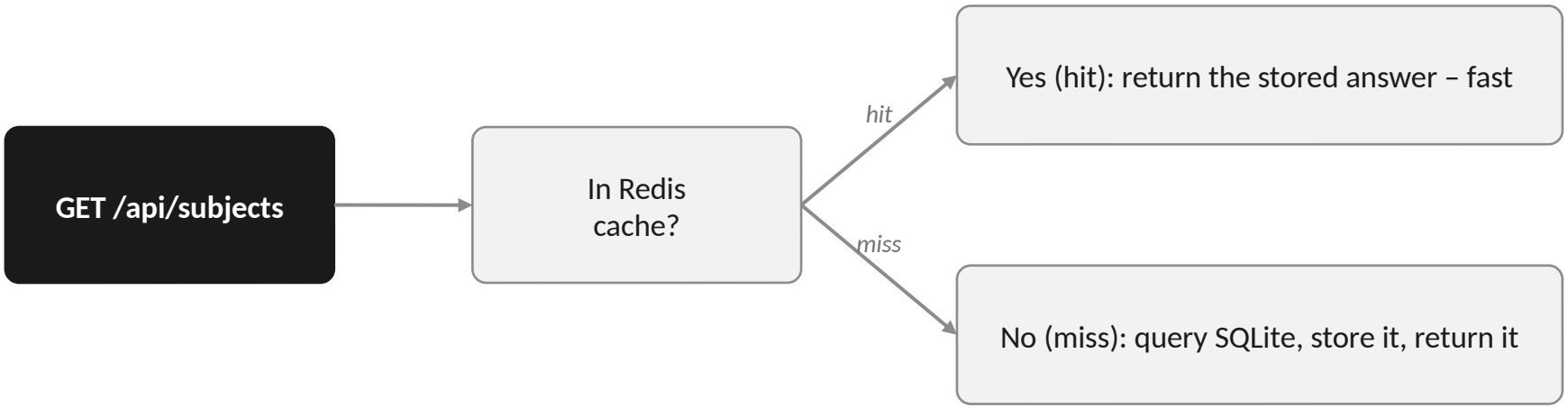
# Today's plan

| | |
|---|---|
| **10 min** | Recap of the whole stack |
| **35 min** | Caching: cached, memoize, invalidation, expiry |
| **30 min** | Summary API + ApexCharts component |
| **20 min** | Security and quality checklist |
| **25 min** | Full end-to-end demo, questions |

## By the end you can

- Cache API responses and explain expiry and invalidation
- Draw charts with ApexCharts in a Vue component
- Walk through the whole app end to end

# Cache hit vs cache miss




Expiry: every entry deletes itself after N seconds. Invalidation: after a write we delete the old copy ourselves.

# What we cache

| Data | How | Cleared when |
|---|---|---|
| GET /api/subjects (same for everyone) | `@cache.cached(key_prefix="subjects")` | any POST / PUT / DELETE in resources.py |
| Quiz list (admin vs user) | `@cache.memoize() on quiz_list()` | same after_request hook |
| GET /api/admin/summary | `@cache.cached(timeout=60)` | expires after 60 s |
| User summary (per user) | `@cache.memoize(timeout=60)` | expires after 60 s |

# Cache, then clear it after every write

backend/application/api/resources.py

```
@resources_bp.get("/subjects")
@auth_required("token")                # 1st: check the token
@cache.cached(key_prefix="subjects")  # 2nd: cached answer
def list_subjects():
    ...

@resources_bp.after_request
def clear_cache_after_write(response):
    if request.method in ("POST", "PUT", "DELETE") \
            and response.status_code < 400:
        cache.delete("subjects")
        cache.delete_memoized(quiz_list)
    return response
```



**Order matters**

auth above cache – never skip the token check



**Invalidation**

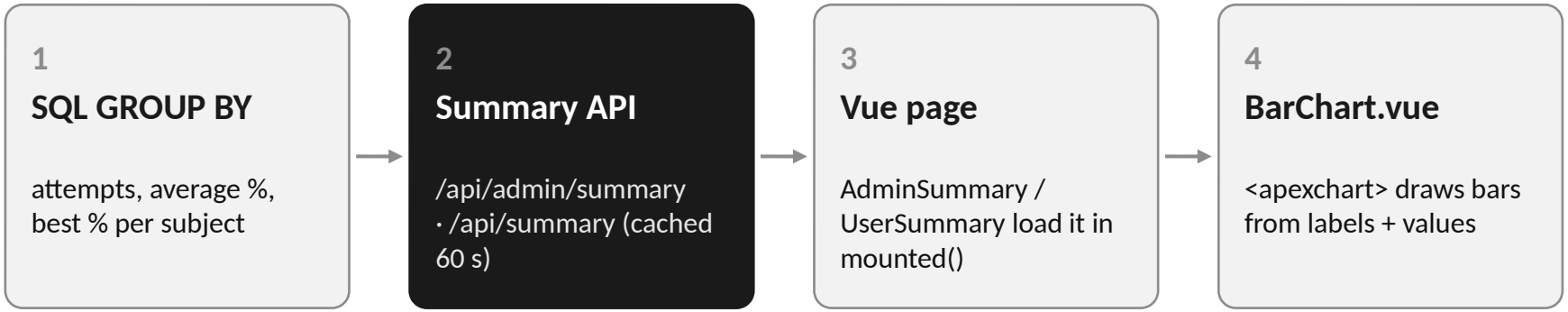
one hook clears both lists after any write



**cache.clear()**

on start, so a DB reset never leaves old data

# From database to chart




npm install apexcharts vue3-apexcharts – one small reusable component draws every chart in the app.

# Register ApexCharts, then a reusable chart

```
// main.js – register once for the whole app
import VueApexCharts from 'vue3-apexcharts'
app.use(VueApexCharts)          // <apexchart> works everywhere

// components/BarChart.vue
<template>
  <apexchart type="bar" height="280"
             :options="options" :series="series" />
</template>
<script>
export default {
  props: { labels: Array, values: Array, max: Number },
  computed: {
    series() { return [{ name: 'Value', data: this.values }] },
    options() {
      return { xaxis: { categories: this.labels },
               yaxis: { min: 0, max: this.max } }
    },
  },
}
</script>
```



**app.use()**

registers <apexchart> once in main.js – no import in each component



**series + options**

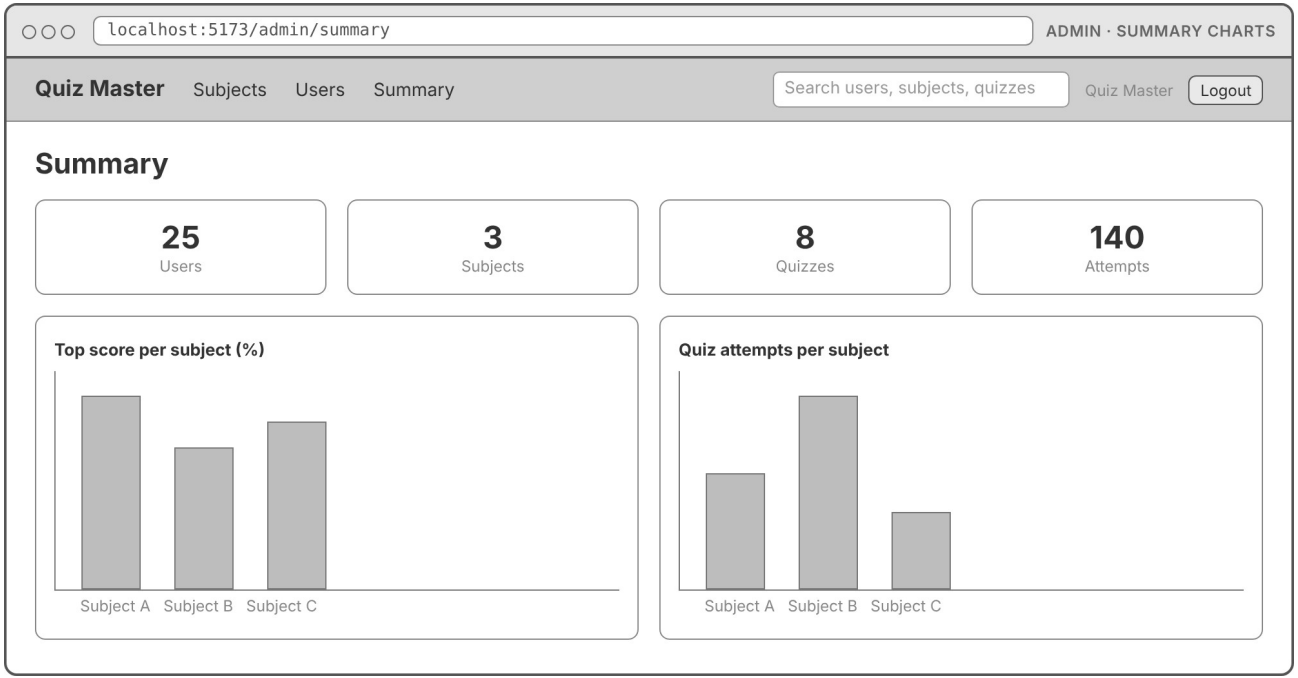
the numbers and the look, both computed from props



**Redraws itself**

new data from the API → computed changes → chart updates

# Admin summary screen




Four totals and two charts. Users get the same idea for their own scores on /summary.

# Security & quality checklist

### Passwords

bcrypt hashes; never returned by any API

### Every API

token check; admin APIs also check the role

### Answers

never sent to users; scored on the server

### Validation

HTML5 in Vue + checks in every Flask route

### SQL

SQLAlchemy queries are parameterised

### Before deploy

change SECRET_KEY, salt and admin password

# Run the finished app

Terminal

```
redis-server
python app.py
celery -A worker.celery worker --loglevel=info
celery -A worker.celery beat --loglevel=info
npm run dev
```

## Check that...

- Admin: create content, search, block a user
- User: take a quiz, see scores and charts
- Export CSV and run both scheduled jobs
- Second load of /api/subjects comes from cache
- UI works at phone width

# Recap

- Redis caching with expiry and invalidation keeps the API fast
- BarChart.vue turns summary APIs into charts for admin and user
- Security checklist: hashing, tokens, roles, server-side scoring, validation
- Quiz Master now covers every core and optional requirement

**Next:** Days 8–10 – doubt-clearing sessions. Send your doubts the day before.